

IMDb 2024 Movie Recommendation System

This project builds a movie recommendation system using IMDb 2024 movie data.

The system collects movie names and storylines, processes the text using Natural Language Processing (NLP), and recommends similar movies based on storyline similarity.

Project Files

- `imdb_2024_movies.csv` – Raw scraped movie data
- `cleaned_movies.csv` – Cleaned and processed dataset
- `Data_Scraping_n_Processing.ipynb` – Notebook used for scraping and preprocessing
- `Movie_Recommendation.py` – Recommendation engine script
- `tfidf_model.pkl` – Saved TF-IDF model
- `tfidf_matrix.pkl` – Saved TF-IDF matrix
- `cosine_sim.pkl` – Precomputed cosine similarity matrix

Step 1: Data Collection

Movie titles and storylines were collected from IMDb and saved into a CSV file.

Step 2: Data Cleaning & Preprocessing

Text cleaning included removing punctuation, stopwords, and unwanted symbols.

The cleaned data was stored in `cleaned_movies.csv`.

Step 3: Text Vectorization

TF-IDF vectorization was applied to convert storylines into numeric vectors.

The trained model and matrix were saved for reuse.

Step 4: Similarity Calculation

Cosine similarity was computed between movie storylines and saved for fast recommendations.

Step 5: Recommendation System

The Python script loads the similarity matrix and recommends the top 5 similar

movies based on a storyline or selected movie.

Business Benefits

- Helps users discover similar movies quickly
- Provides personalized entertainment suggestions
- Useful for OTT and content discovery platforms

Technologies Used

Python, Pandas, Scikit-learn, NLTK, Selenium, Streamlit (optional UI)

Conclusion

This project demonstrates how NLP and machine learning techniques can be used to build an efficient movie recommendation system.